

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The relationship between two variables, x and y , is linear. For every increase in the value of x by 1 , the value of y increases by 8 . When the value of x is 2 , the value of y is 18 . Which equation represents this relationship?

Answer

- A. $y = 2x + 18$
- B. $y = 2x + 8$
- C. $y = 8x + 2$
- D. $y = 3x + 26$

Correct Answer: C

Rationale

Choice C is correct. It's given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the relationship in the xy-plane. It's given that for every increase in the value of x by 1 , the value of y increases by 8 . The slope of a line can be expressed as the change in y over the change in x . Thus, the slope, m , of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8 . Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2 , the value of y is 18 . Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 8(2) + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 2 , not 8 , and when the value of x is 2 , the value of y is 22 , not 18 .

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 2 , not 8 , and when the value of x is 2 , the value of y is 12 , not 18 .

Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 3 , not 8 , and when the value of x is 2 , the value of y is 32 , not 18 .